Pre-Algebra — Course Syllabus

Mary, Star of the Sea School · SY 2026-2027

Mary, Star of the Sea School Mission

Nurture, **U**nite, **I**nspire. We are committed to nurturing each child's curiosity, uniting our classroom in respect and collaboration, and inspiring every student to grow in knowledge, faith, and wonder for the world around them.

Course Description

Math 7 is pre-algebra — the bridge from arithmetic into the symbolic reasoning of high-school math. Topics span proportional reasoning, percent error, scientific notation with negative exponents, multi-step equations, the surface-area-to-volume ratio that explains why cells stay small, and the probability that drives genetics. Each quarter is timed so the math arrives right before STEM (science) class needs it. Expect hands-on labs, classroom studies, project-based assessments, and direct application to Hawaiian contexts — watersheds, microscopic worlds, outbreak modeling, and the Hawaiian monk seal.

Textbook: Glencoe Math Accelerated Pre-Algebra

Teacher

Dr. Peter Park

Email: ppark@starofthesea.org

Phone: (808) 228-2285

Parents may email me or leave a message with the school. I respond within 24 hours.

Materials

- Composition notebook
- Pencils, erasers, highlighters
- Notebook paper
- Calculator
- Folder for handouts
- Textbook (Glencoe Math Accelerated Pre-Algebra)

Grading Policy

20%	25%	30%	25%
Participation	**Tests**	**Homework**	**Quizzes**
Composition books, notes, in-class behavior, supplies	Q&A, written essay (1 per chapter, 20 pts)	Assignments, classwork, class prep, projects	2 per chapter, 5 pts each

Make-ups: Students who score below 80% on tests may make corrections or retest to improve their grade.

Late work: Homework is due the next class meeting. Late work earns a maximum of 50% credit; if not turned in after two class meetings, the grade is a zero.

Course Outline by Quarter — Pre-Algebra

Q1 — Lab Math Refresher & Foundations

- Percent Error with accuracy and precision
- Scientific Notation with negative exponents (microscopic measurements)
- Unit Conversions; rate conversions (km/h → m/s)
- Proportional Relationships; Ratios & Unit Rates

Q2 — Brain Data + Tools for Cells

- Percent Change & Statistics applied to screen-time and memory data — for the STEM Tech & Perception Essay and Brain/Memory Inquiry

- Sampling, Bias & Survey Design
- Rational Number Operations (add/sub/mult/div)
- Algebraic Expressions; Multi-Step Equations
- Rearranging Formulas (D = m/V)
- Surface Area : Volume Ratio — finishes in December, ready for STEM cell unit in January

Q3 — Cells, Geometry & Inequalities

- Geometric Shapes & Triangles; Circles
- Area of Polygons; Surface Area & Volume of 3D Figures
- Angle Relationships; Inequalities
- Percent Change applied — mass changes in the STEM Osmosis Lab
- Probability intro & Exponential Growth — outbreak math for the STEM Virus Debate

Q4 — Probability for Genetics

- Probability & Punnett Squares — lands in early April for STEM genetics unit
- Population Growth Models
- Pharmacokinetics — Drug Half-Life
- Statistics & Sampling; Scale Drawings; Biodiversity Index; Word Problems

---

Student & Parent Communication

Beehively is the platform where students and parents can regularly check grade progress and receive notifications of missing or incomplete assignments.

Parents are contacted by email up to two times if their child is receiving a deficiency, plus standard email notifications. Students are always informed of any grade that is a C or lower and ultimately are responsible for their grades.

Student work is updated regularly.

Middle School Math Goal

From 6th through 8th grade, we focus on more than just numbers — we work to build strong math skills, boost confidence, and develop critical problem-solving abilities that students can apply in every subject and in everyday life. Our approach combines hands-on activities, real-world applications, and a supportive learning environment where students feel encouraged to take risks, ask questions, and think creatively.

By the time they graduate, our students are not only ready for Algebra and beyond, but they also possess the perseverance, analytical thinking, and resourcefulness they need to thrive in high school and in life.